PERCH: Perception–Evidence–Reinforcement CHain for Open-Vocabulary Aerial Object-to-Viewpoint Navigation

Anonymous submission

Abstract

Aerial vision-and-language navigation and object-goal navigation declare success when the agent gets close to the target, yet many UAV missions end with the vehicle keeping watch, so the useful endpoint is an observation pose from which the target is visible, recognizably sized, and safely observed. We keep the standard SR/SPL metric family and replace its proximity-only predicate with a viewpoint-conditioned version (V-SR/V-SPL) computed automatically from simulator ground truth; under this predicate, proximity success overestimates task usability, with SR–V-SR gaps up to 0.44 in closed-loop scenes and 0.73 on photorealistic replays. We propose PERCH, the **P**erception–**E**vidence–**R**einforcement **CH**ain: an occupancy-driven viewpoint affordance module predicts which observation poses can see the target (*map to see*); a VLM-verified candidate belief module gates commitment behind attribute-level verification (*see to trust*); and a viewpoint policy post-trained by group-relative reinforcement learning on verifiable geometric rewards decides when to halt (*trust to stop*). Across four closed-loop AirSim environments and two ground-truth-replay datasets, each module repairs a distinct failure mode; the full stack improves V-SR and V-SPL over both Base-DG and Base-FE on every dataset, and improves proximity SR over Base-DG throughout, with V-SPL rising from 0.675 to 0.803 on closed-loop CityEnviron and from 0.395 to 0.816 on TartanAir's distractor-dense forests. Deployed on an edge multirotor, PERCH stops at usable viewpoints and holds watch over real outdoor targets.

Introduction

A UAV sent to "the white car near the warehouse entrance" has not finished its job when it is merely *near* the car: inspection, patrol, and search expect it to settle at a pose from which the correct instance stays in view at a recognizable scale and a safe standoff. Aerial vision-and-language navigation (VLN) and object-goal navigation (ObjectNav) have made rapid progress on *reaching* language-specified targets (Liu et al. 2023; Lee et al. 2025; Xiao et al. 2025; Xia et al. 2026), but their success criterion, stopping within a distance threshold, counts a terminal frame taken from behind a tree as success. Occlusion, pixel ratio, and clearance all depend on where the vehicle is, so the terminal decision is a full observation pose $v = (x, y, z, \psi)$, raising three questions proximity never asks: *where to look* so the target is unoccluded, *whom to trust* among open-vocabulary look-alikes, and *when to stop* so the terminal pose is worth holding. Next-best-view selection, inspection planning, and perception-aware exploration have long optimized observation poses (Jiang et al. 2026; Jin et al. 2023; Wu et al. 2025; Maboudi et al. 2023; Huang, Qin, and Xue 2026; Placed et al. 2023), yet always for targets already localized by prior meshes or tracking; aerial VLN and ObjectNav solve open-vocabulary grounding but score it blind to observability (Xiao et al. 2025; Wei et al. 2024). Neither community measures, or repairs, the failure between them.

We first make the failure measurable. Keeping the standard SR/SPL family (Anderson et al. 2018), we replace the proximity-only success *predicate*: an episode succeeds under **V-SR** when the terminal pose lies in an observation band $[d_{\min}, d_{\max}]$ around the correct instance, its visible-mask ratio and pixel footprint clear recognizability thresholds (Lin et al. 2014), and the trajectory is collision-free; every term comes from simulator ground truth without human annotation, generalizing the ObjectNav criterion that already demands visibility at the stop (Batra et al. 2020). The gap it exposes is systematic: strong approach agents reach proximity SR of 0.885 on mountainous terrain while only 0.450 of episodes end at a usable viewpoint.

We propose **PERCH**, the **P**erception–**E**vidence–**R**einforcement **CH**ain, whose name states the intended endpoint: the vehicle perches at a viewpoint it can hold and keep watch from. Three plug-and-play modules realize one causal chain: a viewpoint cannot be verified until visible (*map to see*), and stopping cannot be optimal until verification is trusted (*see to trust*, *trust to stop*).

Our contributions are threefold:

- **Map to see.** The *Occupancy-Driven Viewpoint Affordance* module (OVA) back-projects scale-anchored monocular depth into a coarse egocentric occupancy grid and ray-casts hypothesized observation poses into a *viewpoint affordance field* (visible, occluded, unknown), answering where seeing is possible before position is spent. OVA repairs the *invisible-endpoint* failure.
- **See to trust.** The *VLM-Verified Candidate Belief* module (VCB) maintains cross-frame open-vocabulary can-

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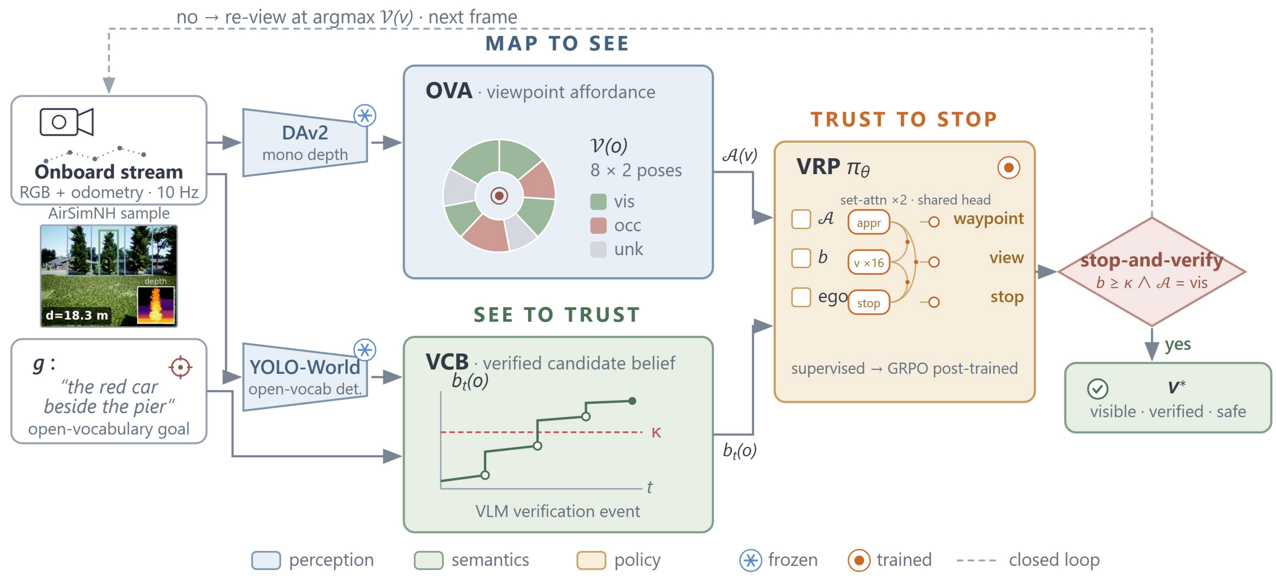

Figure 1: The PERCH inference loop. The front-end lifts each frame into open-vocabulary candidates and depth. **OVA** accumulates depth into an egocentric occupancy grid and ray-casts observation poses into the affordance field $\mathcal{A}(v) \in \{\text{VIS}, \text{OCC}, \text{UNK}\}$; **VCB** maintains cross-frame candidate beliefs $b_t(o)$ refreshed by low-frequency attribute-level VLM queries; **VRP** consumes $(\mathcal{A}, b_t)$ and outputs the next waypoint or Stop. The stop-and-verify gate re-enters the loop on failed verification, so commitment happens only at a visible, verified, safe pose.

didates by spatial re-identification and gates commitment behind low-frequency attribute-level VLM verification: a candidate that fails verification is never committed. VCB repairs *distractor confusion* on qualified instructions.

- **Trust to stop.** The *Verifiable-Reward Viewpoint Policy* module (VRP), a lightweight set-attention policy over the affordance field and belief state, is supervised-pretrained on automatic feasibility labels and post-trained with group-relative policy optimization (Shao et al. 2024) on rewards judged programmatically from simulator geometry (Lambert et al. 2024; Guo et al. 2025). VRP repairs *suboptimal stopping* without a learned reward model.

Across four closed-loop AirSim environments and two ground-truth-replay datasets, the modules form a monotone ablation ladder, and PERCH improves V-SR/V-SPL over two distinct bases; the same stack, on an edge multirotor, perches and holds watch over real outdoor targets.

Related Work

Language-Guided Aerial Navigation

Aerial VLN provides instructions and city-scale scenes (Liu et al. 2023; Lee et al. 2025; Wang et al. 2025; Gao et al. 2025), aerial ObjectNav replaces step-by-step instructions with semantic goals (Xiao et al. 2025), and recent agents add hierarchical planning and geospatial reasoning (Zhang et al. 2025; Xu et al. 2025), edge-deployable scene graphs (Gai et al. 2026), and open-vocabulary outdoor search (Shah et al. 2026; Wei et al. 2024; Xia et al. 2026). Throughout, success is declared within a proximity radius, a criterion inherited from ground VLN and at odds with the ObjectNav canon of visibility from the stop (Batra et al. 2020; Anderson et al. 2018). We keep this literature's metrics, bases, and perception stack (Cheng et al. 2024; Liu et al. 2024; Yang et al. 2024), and change exactly one thing: the success predicate becomes viewpoint-conditioned, exposing an overestimation that our modules then repair.

Active Viewpoint Acquisition and Verifiable Decision-Making

Next-best-view and inspection planning optimize observation poses against known geometry (Jiang et al. 2026; Jin et al. 2023; Huang, Qin, and Xue 2026; Maboudi et al. 2023; Wu et al. 2025; Placed et al. 2023), but assume the target is already found. Dense occupancy prediction from cameras is standard in driving (Mescheder et al. 2019; Huang et al. 2023; Wei et al. 2023; Zhang et al. 2024), yet has not been used to score hypothetical observation poses for an undetermined target. Test-time verification re-ranks or vetoes candidate decisions (Li et al. 2026; Singhi et al. 2026; Liao et al. 2024), and RL with verifiable rewards replaces learned reward models with programmatic checks (Lambert et al. 2024; Shao et al. 2024; Guo et al. 2025; Snell et al. 2024). PERCH composes the three into one closed loop—occupancy scores where seeing is possible, verification decides whom to trust, a verifiable-reward policy decides when to stop—each re-instantiated under aerial constraints: depth arrives without metric scale, distractors share the target's class, and the loop must fit an edge computer.

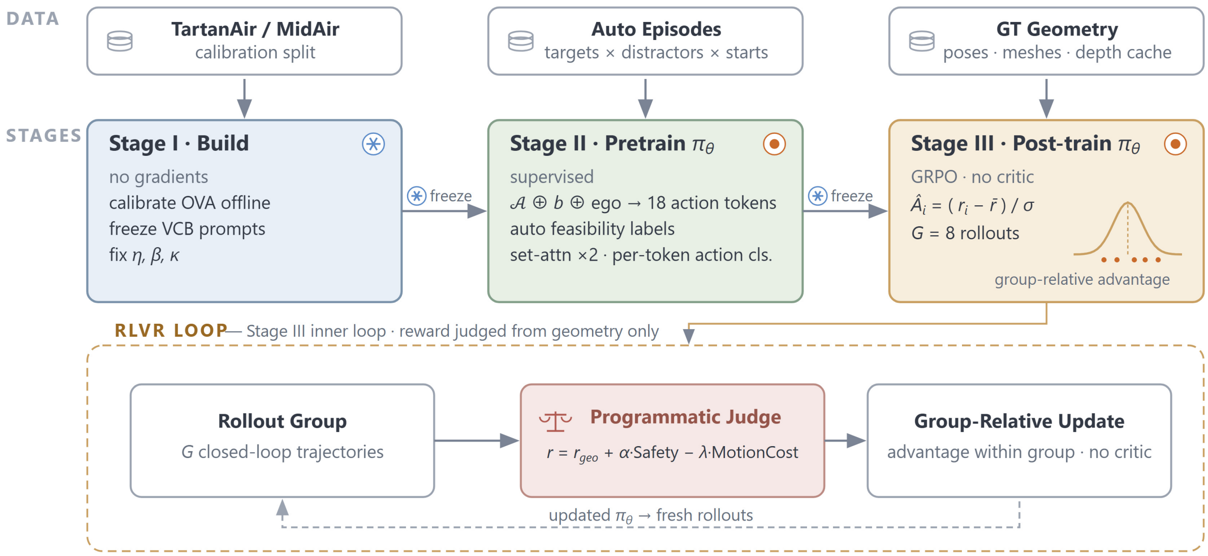

Figure 2: PERCH training pipeline; episodes and viewpoint-success labels are generated programmatically, with no human annotation and no learned reward model. **Stage I** builds OVA and VCB without gradient training; **Stage II** supervised-pretrains VRP on automatically labeled rollouts; **Stage III** post-trains VRP with GRPO-style group-relative advantages on $r = r_{\text{geo}} + \alpha\,\text{Safety} - \lambda\,\text{MotionCost}$.

Method

PERCH composes three modules, OVA, VCB, and VRP, into the closed loop of Fig. 1. Given goal description g and onboard stream $x_{0:T}$ with flight-controller odometry, the agent outputs a terminal pose

$$v^* = \big(\underbrace{\pi_\theta}_{\text{VRP}} \circ \underbrace{\mathcal{B}}_{\text{VCB}} \circ \underbrace{\mathcal{A}}_{\text{OVA}} \circ \underbrace{\mathcal{F}}_{\text{perception}}\big)(x_{0:T}, g), \tag{1}$$

where $\mathcal{F}$ is the shared open-vocabulary detection and monocular depth front-end (Cheng et al. 2024; Yang et al. 2024), $\mathcal{A}$ maps accumulated geometry to a viewpoint affordance field (*map to see*), $\mathcal{B}$ maintains verified candidate beliefs (*see to trust*), and π_θ selects waypoints and the stopping moment (*trust to stop*). Each subsection instantiates one operator; the experiments switch them on in the same order.

Problem Formulation

Given a free-form goal g (category, attributes, spatial relations), the agent starts from a random pose and must terminate at an observation pose $v^* = (x, y, z, \psi)$ such that the referred instance is correct, visible, recognizably sized, and safely observed. We evaluate with the standard family (Anderson et al. 2018) plus the viewpoint-conditioned extension:

$$\begin{aligned}\text{V-SR}: \quad & d(v^*, o^\star) \in [d_{\min}, d_{\max}] \;\wedge\; \text{vis}(o^\star \mid v^*) \geq \tau \\ & \wedge\; \text{px}(o^\star \mid v^*) \geq p_{\min} \;\wedge\; \text{collision-free},\end{aligned} \tag{2}$$

where vis is the ground-truth visible-mask ratio and px the pixel footprint (Lin et al. 2014); V-SPL re-weights V-SR by path efficiency exactly as SPL does SR. V-SR reduces to plain SR when the observation band matches the proximity criterion and the visibility and pixel thresholds vanish, so the two families remain directly comparable. Every term is judged from simulator ground truth while the method may only consume onboard estimates. The difference SR − V-SR is the paper's motivating success-rate discrepancy between arriving and ending at a usable observation pose.

Occupancy-Driven Viewpoint Affordance: Map to See

OVA realizes the operator $\mathcal{A}$ of Eq. (1) and answers *where to look*. In flight, per-frame relative depth is scale-anchored by sparse odometry-consistent landmarks and back-projected into an egocentric occupancy grid M_t with 0.5–1 m voxels, deliberately coarse because the decision it supports is feasibility rather than reconstruction (Zhang et al. 2024). For each candidate o and each pose v in a cylindrical proposal set $\mathcal{V}(o)$, OVA ray-casts $v \to o$ through M_t and emits the *viewpoint affordance field*

$$\mathcal{A}(v) = \big(\underbrace{a_{\text{vis}}(v)}_{\text{VIS/OCC/UNK}}, \underbrace{s_{\text{free}}(v)}_{\text{free-space sector}}\big), \tag{3}$$

where rays through free voxels yield VIS, rays hitting occupied voxels OCC, and rays dominated by unobserved voxels UNK, penalized conservatively so the agent never relies on geometry it has not seen. The field is recomputed as M_t grows, pruning occluded proposals early and bending the approach toward observable sectors; OVA reads no simulator geometry, only onboard estimates.

VLM-Verified Candidate Belief: See to Trust

VCB realizes the operator $\mathcal{B}$ of Eq. (1) and answers *whom to trust*. Open-vocabulary detectors propose generously, and among look-alike instances the top score is routinely wrong. Detections are associated by projected-position gating into persistent candidates o with accumulated belief

$$b_t(o) \;=\; (1-\eta)\, b_{t-1}(o) \;+\; \eta \left[c_t(o) + \beta\, u_t(o)\right], \tag{4}$$

where c_t is detection confidence and $u_t \in \{-1, 0, +1\}$ is the outcome of *attribute-level VLM verification*: at low frequency, the current crop of a top candidate is sent to an open-source VLM (Bai et al. 2025) with questions compiled from g ("is this vehicle white?") (Li et al. 2026; Singhi et al. 2026; Liao et al. 2024). Commitment is gated by *stop-and-verify*: a candidate whose belief has not cleared the bar cannot be committed, and on failure the agent re-views from a nearby affordance-positive pose instead of terminating.

Verifiable-Reward Viewpoint Policy: Trust to Stop

VRP realizes the operator π_θ of Eq. (1) and answers *when to stop*. Fixed-standoff stopping on the detection bearing is exactly what parks agents behind occluders and inside unsafe clearances. VRP is a lightweight set-attention policy $\pi_\theta(a_t \,|\, z_t)$: the fused state $z_t = (\mathcal{A}(\cdot),\, b_t(\cdot),\, \text{ego})$ is tokenized into one action token per candidate viewpoint of $\mathcal{V}(o)$ plus an approach token and a Stop token, each scored by a shared head after self-attention over the set, permutation-invariant in the candidates. Training follows the pipeline of Fig. 2: supervised pretraining on automatically computed feasibility labels, then GRPO-style post-training (Shao et al. 2024), where for each episode context we sample a group of G rollouts $\{\zeta_i\}$ and update with group-relative advantages

$$\hat{A}_i = \frac{r(\zeta_i) - \text{mean}\{r(\zeta_j)\}_{j=1}^{G}}{\text{std}\{r(\zeta_j)\}_{j=1}^{G}}, \tag{5}$$

$$r = r_{\text{geo}} + \alpha\, \text{Safe} - \lambda\, \text{Cost}, \tag{6}$$

$$r_{\text{geo}} = \mathbf{1}_{\text{id}} \big(\beta_b\, r_{\text{band}} + \beta_v\, r_{\text{vis}} + \beta_p\, r_{\text{px}}\big), \tag{7}$$

with soft band / visibility / pixel scores (β_b=0.4, β_v=β_p=0.3), Safe on clearance, and Cost on motion—all judged from simulator geometry, i.e., verifiable rewards with no learned reward model (Lambert et al. 2024; Guo et al. 2025). The group baseline replaces a learned critic under sparse terminal returns (Schulman et al. 2017). To separate reward form from chain state, we train **DG+VSucc** under the same r without OVA/VCB inputs.

The PERCH Loop

Algorithm 1 assembles Eq. (1). At every step the policy either flies to the most promising affordance-positive pose or stops, and stopping is legal only behind the verification gate.

Algorithm 1 PERCH: the Perception–Evidence–Reinforcement Chain

Input: goal g, stream $x_{0:T}$, odometry
Parameter: policy π_θ (Stages II–III); verification bar κ; proposal set size $|\mathcal{V}|$
Output: terminal observation pose v^*

```
M ← ∅;  b_0 ← uniform
for t = 0, 1, ..., T − 1 do
    (D_t, d_t) ← F(x_t, g);  M ← M ⊕ backproject(d_t)
    A ← raycast(M, V);  b_t ← Eq. (4)                ▷ OVA, VCB
    if π_θ(z_t) = STOP then
        if b_t(ô) ≥ κ and a_vis(v_t) = VIS then
            return v_t                               ▷ stop-and-verify passed
        else
            re-view: fly to argmax_{v∈V(ô)} A(v)     ▷ gate failed
        end if
    else
        execute waypoint π_θ(z_t); continue approach
    end if
end for
return v_T
```

each scene affords (tree, rock, car), with attribute qualifiers wherever same-class distractors demand disambiguation. Episodes are generated automatically (target sampling × distractor placement × start randomization) and judged by Eq. (2) from simulator ground truth. Thresholds instantiate COCO-scale recognizability and the ObjectNav visibility canon, fixed once for every method: proximity radius 20 m, band $d_{\min}$=5 m to $d_{\max}$=30 m, visibility τ=0.3, pixel floor 1024 (384 for the smaller car class); the method ordering is stable under moderate threshold perturbations, and the gap they expose tracks scene structure rather than threshold choice. We report SR, V-SR, and V-SPL (Anderson et al. 2018), with V-SR/V-SPL as the deployment-facing bottom line.

Experiments

Task, Episodes, and Metrics

Each episode instantiates *navigate to a viewpoint of the* $\langle g \rangle$, with g a free-form phrase consumed directly by the detector and the VLM verifier; evaluation covers the categories

Environments

Closed-loop evaluation runs in four AirSim (Shah et al. 2017) environments, suburban (AirSimNH, tree), mountainous (LandscapeMountains, rock), urban (CityEnviron, car), and littoral (Coastline, rock), with 40 episodes × 5 seeds each. TartanAir (Wang et al. 2020) (373 cached episodes) and MidAir (Fonder and Droogenbroeck 2019) (1,486; tree) add ground-truth replay over photorealistic synthetic trajectories: recorded trunks are replayed while the full stack, perception, belief, verification, termination, runs onboard-style on cached DAv2 depth, comparing stopping decisions under identical approach geometry. Train/test episodes are split by scene (per-seed replay test pools ∼94 TartanAir, ∼384 MidAir); all numbers are 5-seed means on held-out scenes.

Compared Methods

Published aerial agents couple their termination rules to bespoke detectors, depth models, and action interfaces; reinstantiating the two dominant paradigms on one shared perception stack (YOLO-World + DAv2) equalizes perception

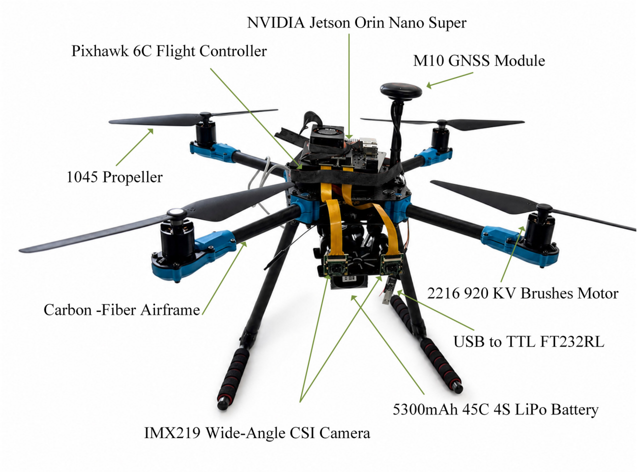

Figure 3: Field platform: a carbon-fiber multirotor (Pixhawk 6C, M10 GNSS, IMX219 wide-angle camera) whose Jetson Orin Nano Super runs the perception-to-policy chain onboard.

across methods, so the tables compare exactly the termination decision under study. **Base-DG** is detect-and-go, flying at the box bearing and stopping at fixed range; **Base-FE** is frontier exploration with open-vocabulary detection and approach (Xiao et al. 2025; Wei et al. 2024). We further compare **Semantic-Nav** (CLIP-guided approach (Radford et al. 2021)), **OVExp**-style exploration (Wei et al. 2024), and two constructed references: **Heuristic-VP** (fixed lateral/height standoff) and **Proximity-Nav** (shortest path to the proximity radius). We also include **DG+VSucc**, which keeps the Base-DG spine and trains a stop head with VRP's verifiable reward while withholding the OVA field and VCB belief from its input. The two references bracket the task: Heuristic-VP applies viewpoint geometry without target-aware reasoning, and Proximity-Nav optimizes arrival alone. All methods consume identical detections, depth, and episode pools. **PERCH** is Base-DG + OVA + VCB + VRP; a swap-base experiment attaches the same modules to Base-FE.

Implementation and Training Data

OVA uses $0.75\,\mathrm{m}$ voxels, a $60\,\mathrm{m}$ egocentric extent, and 16 cylindrical proposals per candidate. VCB queries Qwen2.5-VL-7B (Bai et al. 2025) at most once per 10 frames per candidate with β=0.4, η=0.3, κ=0.6. VRP embeds the 18 action tokens (16 proposals, approach, Stop) through two pre-LN self-attention blocks (width 128, 4 heads, FFN 256; $\sim$0.3M parameters) with a shared per-token scoring head. Stage II pretrains on 350 automatically generated episodes from the training scenes with programmatic feasibility labels; Stage III post-trains with GRPO at group size G=8 over the same contexts (2,800 verifiable-reward rollouts per seed; α=0.3, λ=0.05, 5 seeds). The ablation replays cached depth so every rung sees identical perception.

Results

Table 1 contrasts eight methods across six datasets; Table 2 unfolds PERCH into the cumulative ladder base $\rightarrow$ +OVA $\rightarrow$ +VCB $\rightarrow$ +VRP. PERCH improves Base-DG on every dataset and metric, and improves V-SR and V-SPL over Base-FE throughout, with the largest gains where occlusion and distractors bind: TartanAir V-SPL rises from 0.395 to 0.816 and V-SR from 0.480 to 0.828, CityEnviron V-SPL from 0.675 to 0.803, and macro-averaged over the four AirSim environments PERCH attains SR/V-SR/V-SPL of 0.730/0.535/0.476 against Base-DG's 0.666/0.475/0.406. DG+VSucc, which shares VRP's reward but not the chain state, trails on V-SR and V-SPL on every dataset (0.515 vs. 0.828 on TartanAir): a stop head that observes neither visibility nor identity cannot tell occluded canopy approaches from usable viewpoints. The modules thus buy observability without giving up proximity relative to the Base-DG spine.

Arriving Is Not Observing

On LandscapeMountains, Base-DG arrives on 88.5% of episodes yet only 45.0% terminate at a usable viewpoint; on TartanAir's forest trunks the gap reaches 0.49 (Base-DG) and 0.73 for methods without viewpoint reasoning (Semantic-Nav, Heuristic-VP at 1.00 SR vs. 0.27 V-SR). The gap is no artifact of strict thresholds—apart from the arrival-only Proximity-Nav reference, CityEnviron's open streets show gaps of only 0.02–0.08—but concentrates where occlusion and terrain make observation geometry nontrivial. Conversely, Proximity-Nav collapses V-SR to 0.12–0.48 in three of four closed-loop environments, so viewpoint quality is no free by-product of reaching. On replay the recorded trunk fixes approach geometry, so Proximity-Nav, an oracle-style arrival reference, can tie PERCH on V-SR while remaining far behind on V-SPL. The gap thus summarizes decision quality at termination, the quantity a stop-and-watch mission cares about.

Ablation Study

Table 2 switches the modules on cumulatively over Base-DG; identical episode pools and seeds make every rung a paired contrast, and across six datasets the ladder has eighteen opportunities to lower V-SR and takes none. OVA lifts SR and V-SR wherever occlusion binds (+0.040 V-SR on CityEnviron's occluding car rows, +0.020 SR on LandscapeMountains' ridgelines) and stays small but never negative on open terrain, consistent with a veto on infeasible endpoints. VCB jumps largest where distractors are dense (+0.155 V-SR on TartanAir's same-class forests) and stays inert with few look-alikes; a gate that never fires costs nothing. VRP converts the remaining headroom into terminal pose quality: V-SPL jumps to 0.803 on CityEnviron and 0.816 on TartanAir because the policy stops earlier, straighter, and inside the band, and macro V-SPL rises from .441 to .476 in this rung, consistent with stopping quality being the last unaddressed failure once visibility and identity are handled. Fig. 4 illustrates the repaired failure modes.

The hardest regimes preserve the ordering: on MidAir's photorealistic flight replays, PERCH attains tied-best V-SR

	AirSim closed loop												*ground-truth replay*					
	AirSimNH			LandscapeMts			CityEnviron			Coastline			TartanAir			MidAir		
Method	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL
Semantic-Nav	.595	.510	.451	.890	**.500**	**.467**	.924	.887	.740	.415	.200	.175	**1.00**	.270	.182	.900	.845	.820
Base-DG	.575	.485	.420	.885	.450	.390	.845	.805	.675	.360	.160	.140	.970	.480	.395	.870	.835	.810
OVExp	.500	.385	.310	.715	.355	.280	.802	.784	.517	.365	.155	.110	**1.00**	.289	.191	.760	.720	.690
Base-FE	.455	.375	.290	.680	.305	.241	.789	.713	.507	.350	.165	.126	**1.00**	.396	.292	.840	.790	.760
Heuristic-VP	.455	.375	.363	.865	.465	.428	.921	.878	.685	.390	**.205**	**.180**	**1.00**	.270	.178	.890	.845	.800
Proximity-Nav	.400	.390	.390	.380	.120	.120	.887	.484	.452	.310	.155	.154	.998	**.828**	.567	.910	**.888**	.828
DG+VSucc	.585	.490	.430	.890	.455	.395	.860	.820	.700	.365	.165	.145	.980	.515	.430	.880	.845	.820
PERCH (Ours)	**.630**	**.540**	**.488**	**.934**	.492	.431	**.936**	**.901**	**.803**	**.420**	**.205**	**.180**	.998	**.828**	**.816**	**.920**	**.888**	**.886**

Table 1: Main comparison across six datasets (AirSim: 40 episodes $\times$ 5 seeds per environment; TartanAir/MidAir: 5-seed splits over the cached replay episodes; goal classes: tree/rock/car/rock/tree/tree). **SR**: proximity success; **V-SR**: viewpoint-conditioned success (Eq. (2)); **V-SPL**: path-efficiency-weighted V-SR. **DG+VSucc**: Base-DG with a stop head trained under VRP's verifiable reward but without OVA/VCB inputs. All entries are 5-seed means over identical per-seed episode pools (per-cell std ≤ 0.045). Best per column in **bold**.

	AirSimNH			LandscapeMts			CityEnviron			Coastline			TartanAir			MidAir		
Stage	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL	SR	V-SR	V-SPL
base (Base-DG)	.575	.485	.420	.885	.450	.390	.845	.805	.675	.360	.160	.140	.970	.480	.395	.870	.835	.810
+ OVA	.595	.505	.440	.905	.465	.410	.870	.845	.700	.380	.175	.155	.980	.500	.415	.885	.850	.825
+ VCB	.610	.520	.455	.920	.475	.420	.900	.875	.720	.400	.190	.168	.990	.655	.605	.900	.865	.850
+ VRP (PERCH)	.630	.540	.488	.934	.492	.431	.936	.901	.803	.420	.205	.180	.998	.828	.816	.920	.888	.886

Table 2: Cumulative ablation ladder on the Base-DG spine (5-seed means). Each row switches on one module in the causal order perception $\rightarrow$ semantics $\rightarrow$ policy. V-SR and V-SPL never decrease at any rung on any dataset; macro-averaged AirSim V-SR climbs $.475 \rightarrow .498 \rightarrow .515 \rightarrow .535$.

(0.888) and best V-SPL (0.886); on Coastline, where rocks sit near the detector's visibility floor, it is best or tied-best on every metric; on LandscapeMountains it posts the best SR (0.934) while trailing the best V-SR by 0.008, a margin Semantic-Nav buys at a 0.044 SR cost. No compared method Pareto-dominates PERCH on any dataset.

Real-World Deployment

The mission is only credible if the chain survives real optics, depth error, and wind. The perception-to-policy chain runs onboard the Jetson Orin Nano Super of Fig. 3 at 4–5 Hz (YOLO-World-S detection, DAv2 ViT-S depth, occupancy accumulation, frozen VRP); the verifier's one-query-per-ten-frames duty cycle makes it a light service. We flew 20 field episodes over five park scenes (Fig. 5), each naming a tree target among dense same-class distractors from launch points 25–60 m away. In every episode the vehicle stopped at a pose that passed the stop-and-verify gate and held it for 60 s of observation, and post-hoc inspection confirmed each committed viewpoint unoccluded, inside the band, and at a safe standoff. VCB rejects same-class distractors in cluttered canopies (red boxes in Fig. 5), and committed poses bend off the detection bearing wherever the line of sight is blocked, zero-shot from the same policy post-trained purely in simulation.

Limitations and Future Work

V-SR uses simulator ground truth for evaluation, with fixed thresholds shared by all methods; mission-specific bands remain future work. PERCH also inherits persistent monocular-depth and open-vocabulary detection errors, although occupancy accumulation and cross-frame verification mitigate transient noise and false positives. Future work will adapt thresholds to instructions and strengthen recovery under perception failures.

Conclusion

Arriving is not observing. Replacing the proximity-only success predicate with a viewpoint-conditioned one shows that approach-paradigm aerial navigation systematically overestimates task usability, and a chain of three modules repairs it: map to see, see to trust, trust to stop. The ladder is monotone on every dataset, the gains transfer across two bases, and the same stack runs on an edge multirotor that stops at usable viewpoints and keeps watch over real targets. Code, the episode generator, and the V-SR evaluation suite will be released.

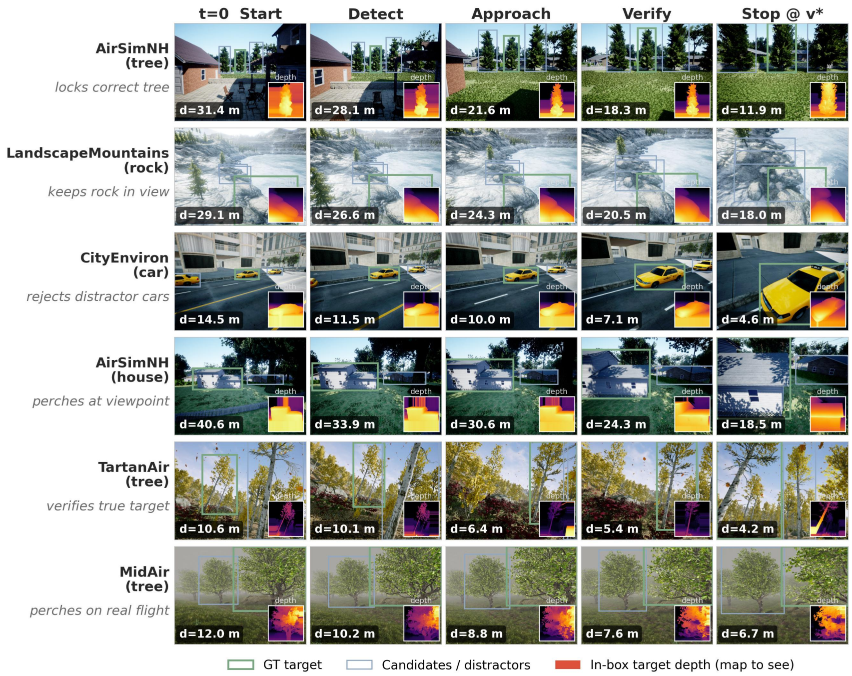

Figure 4: Qualitative episodes across six datasets (rows) over time (columns: start, detect, approach, verify, stop). Green: ground-truth target; blue: candidates/distractors; insets: monocular depth with in-box target depth (map to see). Distances to the target (bottom left) decrease monotonically into the $[d_{\min}, d_{\max}]$ band.

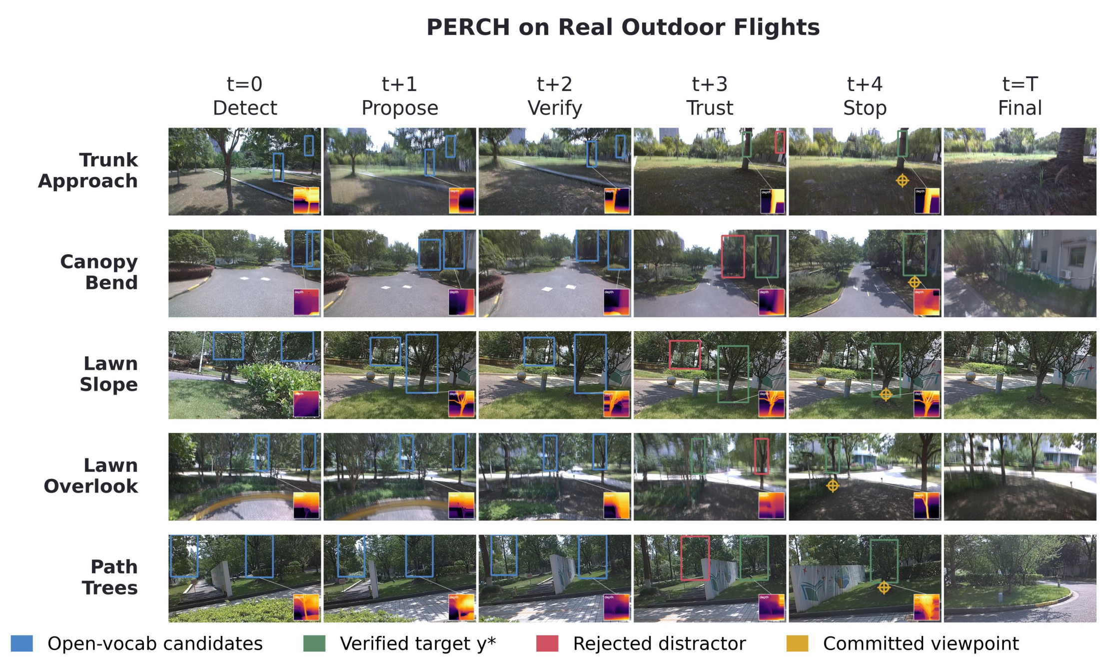

Figure 5: PERCH on real outdoor flights. Rows: five park scenes with dense same-class tree distractors; columns: detect → propose → verify → trust → stop, with onboard depth as insets. Blue: open-vocabulary candidates; green: the target that passed attribute verification; red: rejected distractors; gold: the committed viewpoint where the vehicle holds position and keeps watch.

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